

# Harjot Singh Raith

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## Education

Narsee Monjee Institute of Management and Studies, Navi Mumbai  
*B.Tech in Artificial Intelligence and Data Science*

June 2022 – July 2026  
CGPA: 3.2/4.0

## Research Experience

**eYantra Research Intern at IIT Bombay**

*Embedded Real-Time Systems Laboratory (eRTS Lab), IIT Bombay*

IIT Bombay, India

June 2025 – Present

- Developed a full-stack AI-driven Thematic Analysis Tool for multi-format document processing and automated inductive/deductive coding. Built a FastAPI-PostgreSQL backend with LLM integration to generate codes, themes, and research reports. Designed a React/MUI frontend with interactive visualizations such as Sankey diagrams. Delivered a streamlined qualitative research platform that accelerates coding, theme generation, and insight extraction.

**eYantra Research Intern at IIT Bombay** [Certificate](#)

*Embedded Real-Time Systems Laboratory (eRTS Lab), IIT Bombay*

IIT Bombay, India

May 2024 – July 2024

- Analyzed large-scale user interaction logs from an educational platform using Python, Google OCR, CNN models, and API-based data extraction to assess scaffold availability and effectiveness. Identified resource-access patterns, evaluated dwell time, and applied data mining techniques including clustering, heatmaps, sequence analysis, and predictive modeling to uncover user behavior insights. Developed comprehensive visualizations and pattern-recognition workflows, optimizing scaffold evaluation and enhancing understanding of learner engagement.

## Publication

**Enhancing and analyzing log generation for Collaborative Problem-Solving activities: Video analysis and OCR techniques**

Suprabha Jadhav, Parth Jain, **Harjot Singh Raith**, Sridhar Iyer, Kavi Arya

[Research Paper Link](#)

May 2024 – Dec 2024

**YTubeRAG: Leveraging YouTube Content for Enhanced LLM Training with Optimized Vector Space Retrieval**

**Harjot Singh Raith**, Kartikeya Mudliyar, Ishika Mohan, Aditya Kasar, Sakshi Indolia

[Paper Certificate](#)

Nov 2024 – May 2025

**AI-Driven Timetable Generation and Optimization by Constraint-Based Scheduling**

**Harjot Singh Raith**, Kartikeya Mudliyar, Toral Shah

Accepted

Jan 2025 – Nov 2025

## Projects

**LiveWire : Real time code compiler**

[Github](#) [Website](#) [Youtube](#)

(AWS Global Vibe)

- Built a real-time code-compiler platform for developers using Monaco, WebSockets, and Gemini AI.
- Tools Used: TypeScript, Vite, Tailwind CSS, Monaco Editor, Socket.IO, Express, Node.js, CodeMirror, Vercel, Railway.

**ArgoMind AI : Dashboard and Mapbot of Argo floats.** [Github](#) [Website](#) [Youtube](#)

(SIH'25)

- Developed an automated pipeline to ingest FTP raw files, store them in S3, and generate a sequential schema for real-time dashboards and a map-based RAG system using Argo data with ML-driven ocean disaster predictions.
- Tools Used: React, Python (UV), FastAPI, Xarray, Tailwind CSS, Leaflet.js, AWS S3, AlloyDB, Docker, Gemini API.

**Almanac AI : TimeTable generation and optimization** [Github](#) [Website](#) [Youtube](#)

(Capstone Project)

- Created a GA/CSP/backtracking-based optimized timetable generator with RAG for quick class and faculty lookup.
- Tools Used: React, Vite, Tailwind CSS, MUI, Axios, Node.js, Express.js, MongoDB, Gemini AI, Vercel, Render.

**Biros : Bio-Inspired Robotic Snake**

[Github](#) [Youtube](#)

(Gujarat Robofest'25)

- Created an autonomous robotic snake using CNNs, kinematics, and LiDAR for detection in confined and disaster areas.
- Tools Used: ESP32, Servo Motor MG995, VL53L0X, LiDAR Sensors, OpenCV, Inverse and Forward Kinematics.

## Technical Skills

**Languages:** Python, CUDA, Shell, Latex, C, C++, Java, SQL, Javascript, Typescript.

**Dev Tools:** Linux, AWS, GCP, Docker, Kubernetes, Hugging Face, Wandb, Overleaf, GitHub.

**Frameworks:** TensorFlow, PyTorch, FastAPI, Flask, OpenCV, Langchain, Streamlit, Keras, Selenium.

## Awards

Won Judges' Choice Award out of 503 teams at eYIC 2023-24 Finals, IIT Bombay.

[Certificate](#)

Won 2nd place in VOIS International Hackathon 2024 by Vodafone, held across India, Romania and Egypt

[Certificate](#)

Competed in over 20 hackathons, securing 8 wins and numerous awards.

[All Certificates](#)

## Position and Responsibility

Head of the Robotics Club, NMIMS Navi Mumbai

July 2024 - present

Core member of the Manthan Club, NMIMS Navi Mumbai

September 2022 - Jan 2023